

Unit 11: Web Server Configuration

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Objectives

After studying this unit, you will be able to

- Know what a web server is
- Understand how to set up a web server
- Know how to install a web server on Linux machine
- Understand Apache web server
- Understand the important packages of Apache web server

Introduction

A web server is nothing but software and hardware that uses the Hypertext Transfer Protocol, commonly known as HTTP, and some other protocols that respond to request from clients made on the World Wide Web. The main job that the webserver performs is to display the content of the website, which it does by storing, then processing, and eventually delivering the webpages to the user who has requested it. The web server also supports Simple Mail Transfer Protocol or SMTP and File Transfer Protocol or FTP and HTTP. These are used to transfer files for emailing and even for storage.

The web server hardware gets connected to the internet, allowing the exchange of data with the other devices related to it. The web server software controls how the user assesses the files that have been hosted. The web server process is basically a client and server model example. All the computers that host websites should have the webserver software. Web serves find use in web hosting or the hosting of data for websites as well as for all kinds of web-based applications. The web server software gets accessed using the domain name of a website. This then ensures that the content of the site gets delivered to the user who has requested it. The software part of the webserver is also made up of various components and has at least a single HTTP server. The HTTP server understands the URLs and the HTTP. The web server hardware is basically a computer that will store the webserver software as well as the files that are related to the website. These include documents, HTML, JavaScript files, and images.

Uses of Web Server

The web server is basically a part of a large internet package. It also offers many programs that are related to the intranet. The web server is used to:

- Send and receive emails.
- Download the file transfer protocol or FTP request
- Build and publish webpages.

The basic kinds of web servers can support the scripting on the server-side, which is used to employ the scripts on the webserver. This can be customized as per the request of the client. The server-side scripting works on the server machine, and this comes with a broad feature set that offers access to the database. The server-side scripting makes use of Active Server Pages or ASP, Hypertext Pre-processor or PHP, and many other scripting languages. The process also lets the HTML documents to get created.

Dynamic and Static Web Servers

A web server may be used as static or dynamic content. The static content is the one that is fixed. The static web server contains HTTP software and computer. This is static because the server sends hosted files as is present to the browser.

On the other hand, the dynamic web browser will have the webserver and the software like the database and the applications server. This is dynamic because the application server is used to update the files hosted before these are sent to the browser. The web server generates content when the database requests it. The process is flexible but complicated too.

A web server can host a single website or many websites with the help of the same software and hardware resource. This is called virtual hosting. The answer to what is the role of a web server is here. Web servers are also capable of limiting the speed of the response to various clients, which in turn does not allow a single client to dominate the resources. This is used to satisfy the requests of many clients. The web servers will typically host the websites that are internet accessible. These can also be used to communicate between the web clients and the servers in the local network area. This could be like through the intranet of a company. The web server could be embedded in a device like a digital camera. This lets the users communicate with the device using a commonly available web browser.

Basic Common Features

HTTP: The support for one or more versions of HTTP protocol in order to send versions of HTTP responses compatible with versions of client HTTP requests, e.g., HTTP/1.0, HTTP/1.1 plus, if available, HTTP/2, HTTP/3.

Logging: Usually web servers have also the capability of logging some information, about client requests and server responses, to log files for security and statistical purposes.

Setting up a web server

When we want to publish web pages on the Internet or on an intranet, we use a web server. In essence, a web server is an application that does two things:

- 1) It listens for page requests.
- 2) When it receives a page request, it examines the request and responds with the page that was requested.

Of course, there are many different web browsers in existence (including Mozilla, Opera, Internet Explorer, and others), and there are also a great many types of web server software. To enable a browser to request pages from a web server, they communicate using Hypertext Transfer Protocol (HTTP) – this is the standard protocol for the Internet. The request and response messages are composed using HTTP, and this is what allows any browser to request web pages from any type of web server. By default, all web servers listen for HTTP requests on port 80. Web servers also use port 443 to listen for requests made through secure HTTP connections, over SSL (secure sockets

layer), through a protocol called HTTPS. So, if you want to publish your own web site, you'll need a machine with some web server software.

Why to install a web server on Red Hat Linux Machine

Well, here are two scenarios:

- First, if you're building a web site, then you'll need a web server so that you can test your site as you're developing it.
- Second, although you might not host an Internet site from your own machine, you might host an intranet site – a private web site available only to other machines inside your private network.

Commercial and open-source web servers

Commercial

- Netscape
- Iplanet
- SunONE
- Microsoft
- Zeus

Open Source

- Apache
- Thttpd
- Redhat TUX

11.1 The Apache Web Server

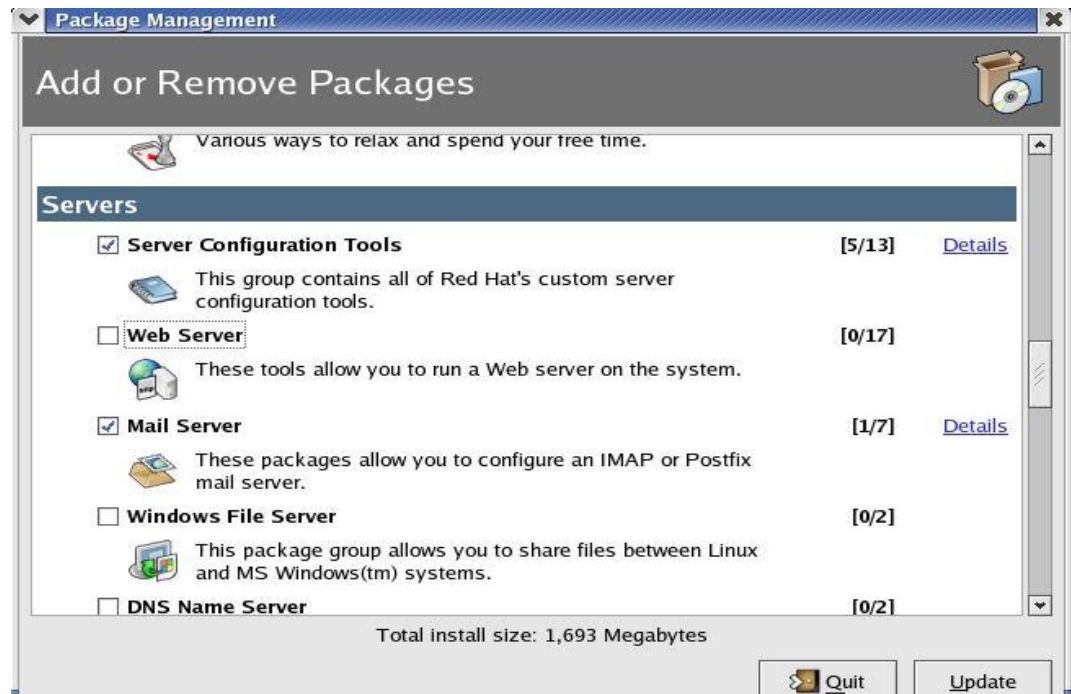
At the time of writing, 66% of all web sites are hosted on Apache web servers, most of them running on Linux or Unix operating systems. Apache's popularity is due not only of its open-source pedigree, but also to its highly competitive levels of performance, functionality, stability, flexibility, and security.

- 1) **Flexibility:** Apache's flexibility comes from the fact that it is a modular web server. That means that you can meet your requirements by plugging any number of external modules into the core httpd daemon. Of course, being open-source software, you also have access to Apache's source code, which you can customize to fit your needs.
- 2) **Scalable & Portable:** Apache is also very scalable. You can run Apache on high-end hardware, and it's possible to increase the capacity of Apache web servers by sharing the load across any number of servers. It's also very portable, being available for several operating systems.
- 3) **Security:** Apache's security is very good in comparison to other web servers. Moreover, the Apache Foundation is extremely active in the continued defense of Apache from security problems – particularly in the form of announcements and patches.
- 4) **Functionality:** Apache performs very well – it boasts a highly optimized daemon for serving static content which dramatically outperforms its nearest rivals. Moreover, it rarely crashes and achieves extremely long up-times.
- 5) **Documentation:** Apache comes with detailed documentation, which helps to make the setup and configuration easy.
- 6) **Support:** There's a wide network of support for Apache, in the form of mailing lists, newsgroups, and commercial vendors like Red Hat.
- 7) **Stability:** Apache development is active. The Apache Foundation is actively involved in development of new modules; new versions of Apache to make it reliable stable and secure.

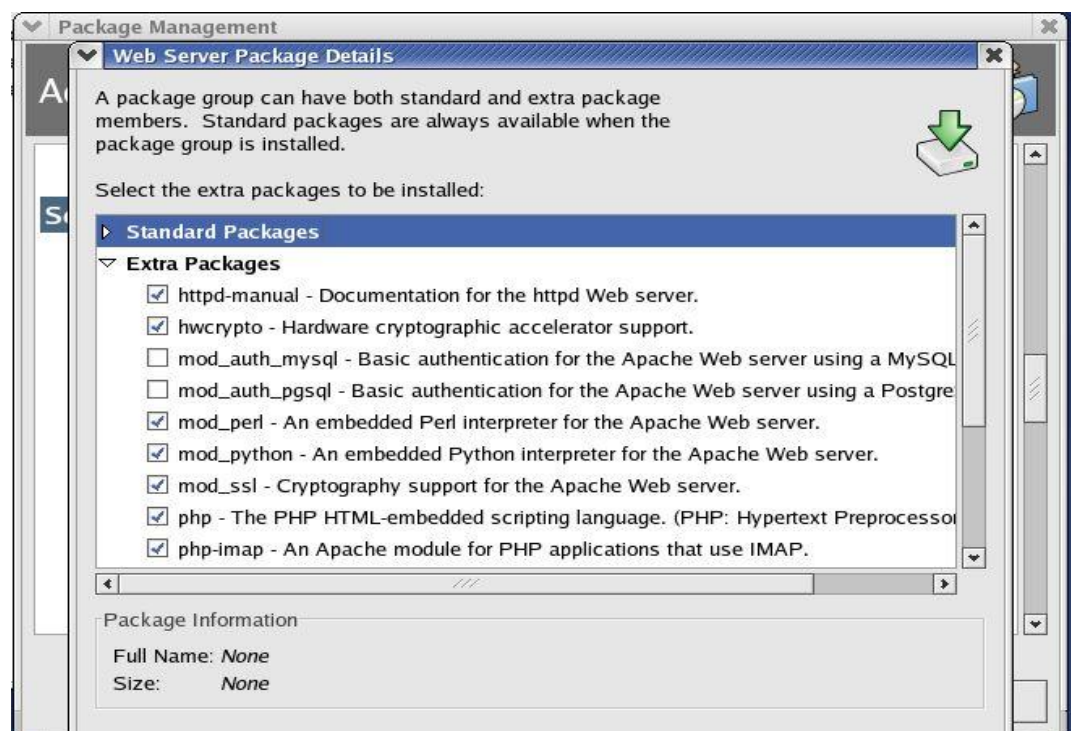
11.2 Installing Apache

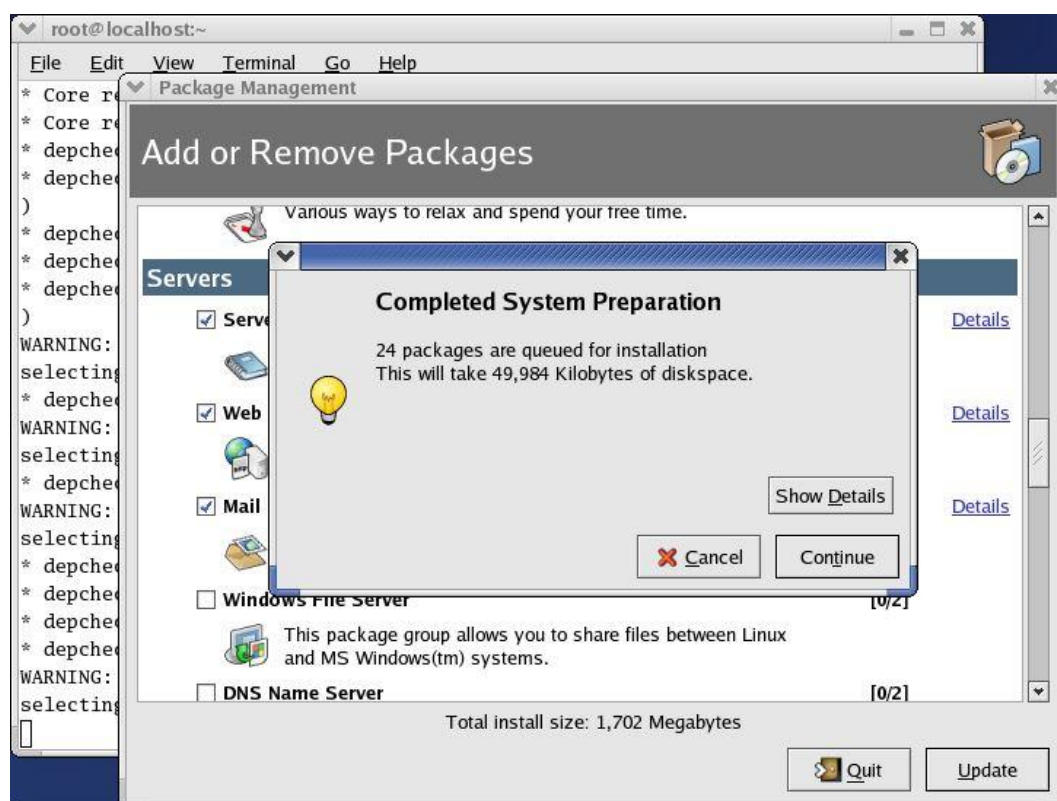
Apache is a modular server – the core server provides the basic functionality, with extended features available in various modules. This makes it very flexible and easy to configure, because you need to configure only the modules you need. So, it's worth looking at how to control the installation and removal of these modules. There are two ways to start RPM's graphical interface:

One is to go to Main Menu | then System Settings | then Add/Remove Applications. Other is to open `$ red hat-config-packages`.



In this we need to search web server package.





Important packages

There are some important packages which needs to be installed for web server. These are:

Package	Description
httpd-manual	Contains the documentation for the Apache web server. After installation, you can access this documentation from the command line by typing <code>man httpd</code> .
hwcrypto	Provides support for hardware SSL acceleration cards. This package should be installed if you have hardware SSL acceleration cards like Ncipher Nforce on your server.
mod_ssl	Provides an SSL interface to the HTTPS web server, and hence enables the Apache web server to support SSL. This package should be installed if you want to provide secure connections to your clients.
Php	Provides the PHP module for Apache, which enables the web server to serve PHP web pages. This package is required if you if you want to host web sites which contain pages written with the PHP scripting language.
webalizer	Provides programs for web server log file analysis. This package enables you to generate HTML usage reports for your website.

Apache Configuration Files

There are various configuration files for configuring the Apache web server in the computer system.

- 1) The `/etc/httpd/httpd.conf` file is Apache's main configuration file.
- 2) The `/etc/httpd/conf.d` directory contains configuration files for any installed modules (such as PHP, SSL, and so on).
- 3) The `/etc/httpd/logs` directory is a symbolic link to `/var/log/httpd` directory, which contains all the Apache log files.
- 4) The `/etc/httpd/modules` directory is a symbolic link to `/usr/lib/httpd/modules` directory, which contains all the Apache modules configured as dynamic shared objects. Dynamic shared objects (or DSOs) are modules that are compiled separately from the Apache httpd binary. They are so-called because they can be loaded on demand.
- 5) The `/etc/httpd/run` directory is a symbolic link to `/var/run`, which contains the process ID file (`httpd.pid`) of the httpd process.
- 6) `/etc/rc.d/init.d/httpd` is a shell script, used for starting and stopping the Apache web server.

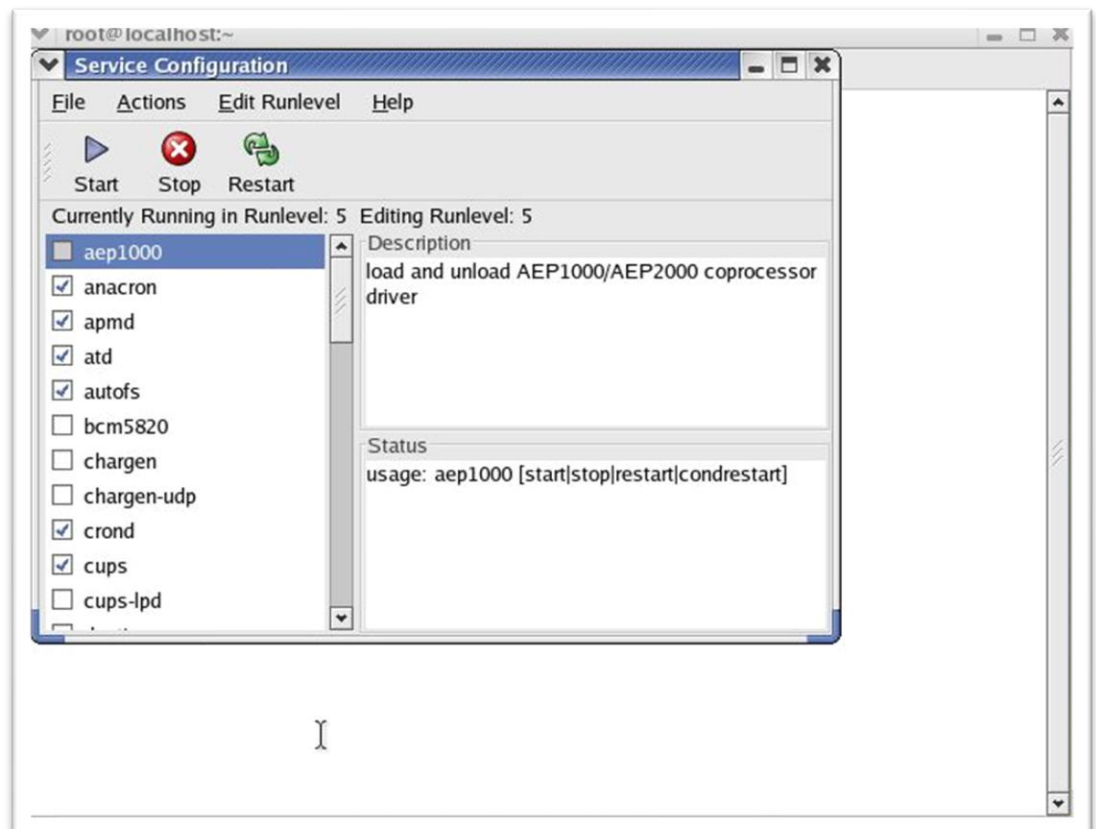
Starting Apache for the First Time

There are two ways by which Apache can be started. These are:

- 1) Main Menu | System Settings | Server Settings | Services
- 2) `$ redhat-config-services`

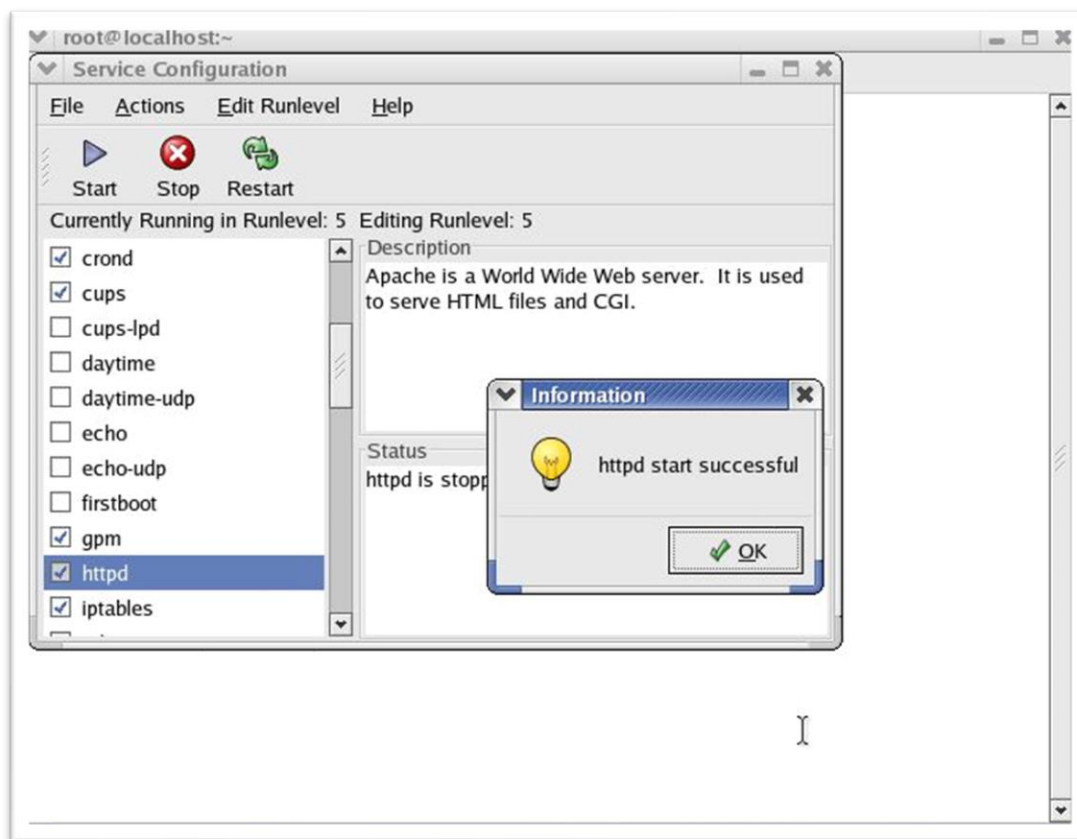
We can opt any of the ways. It would ask for root password if you started as a normal user.

Service Configuration Dialog: Here we need to start the service.

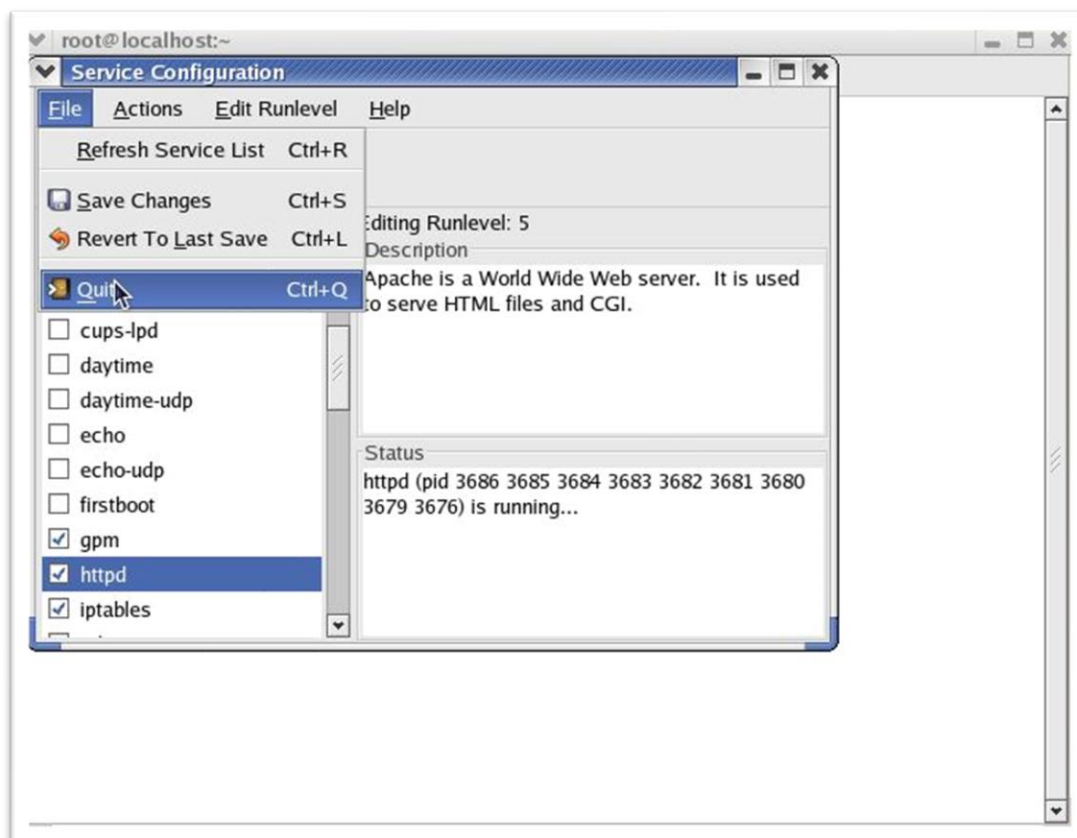


Unit 11: Web Server Configuration

Httpd need to be checked here.



When this is done. We need to save the changes and quit.



To control the Apache web server from the command line, we can use the service command to fire the httpdscript. Here's how we use it to start the web server:

```
# service httpd start
```

Starting httpd: [OK]

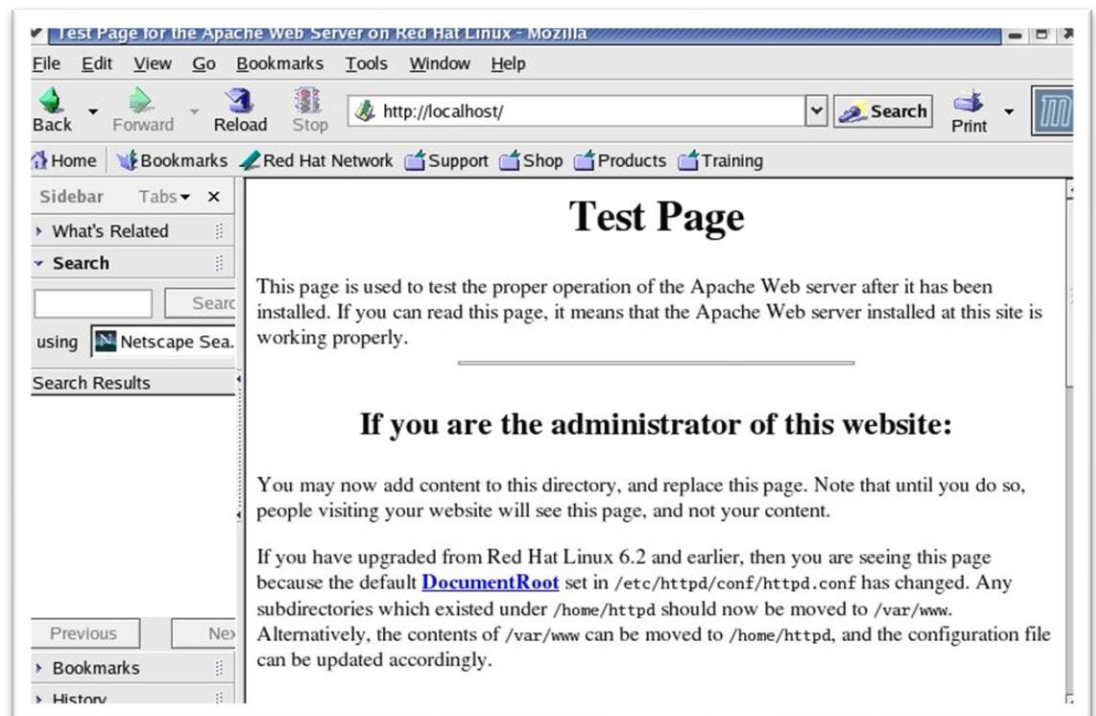
If there are difficulties in starting the web server, then you'll find out about it here. For example, if you attempt to do this without root privileges, then you'll get a message telling you that permission is denied. And here's another example:

```
# service httpd start
```

Starting httpd: httpd: Could not determine the server's fully qualified domain name, using 192.168.0.99 for ServerName [OK]

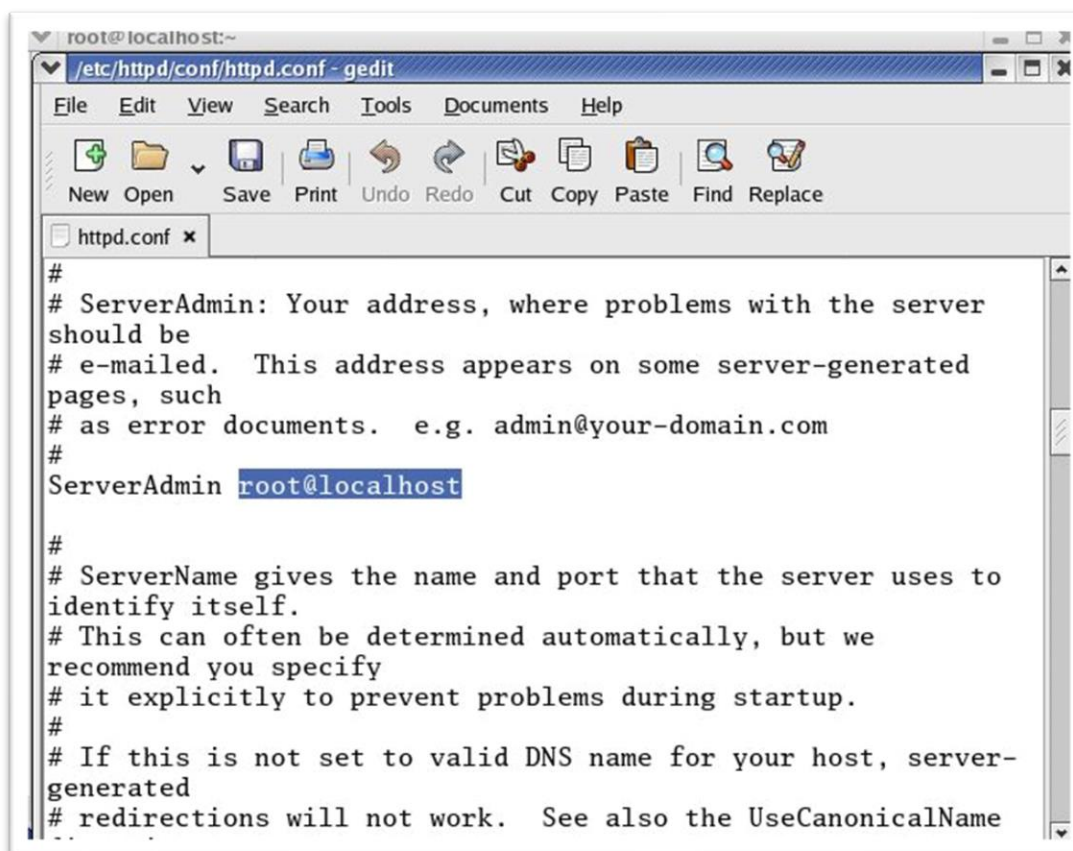
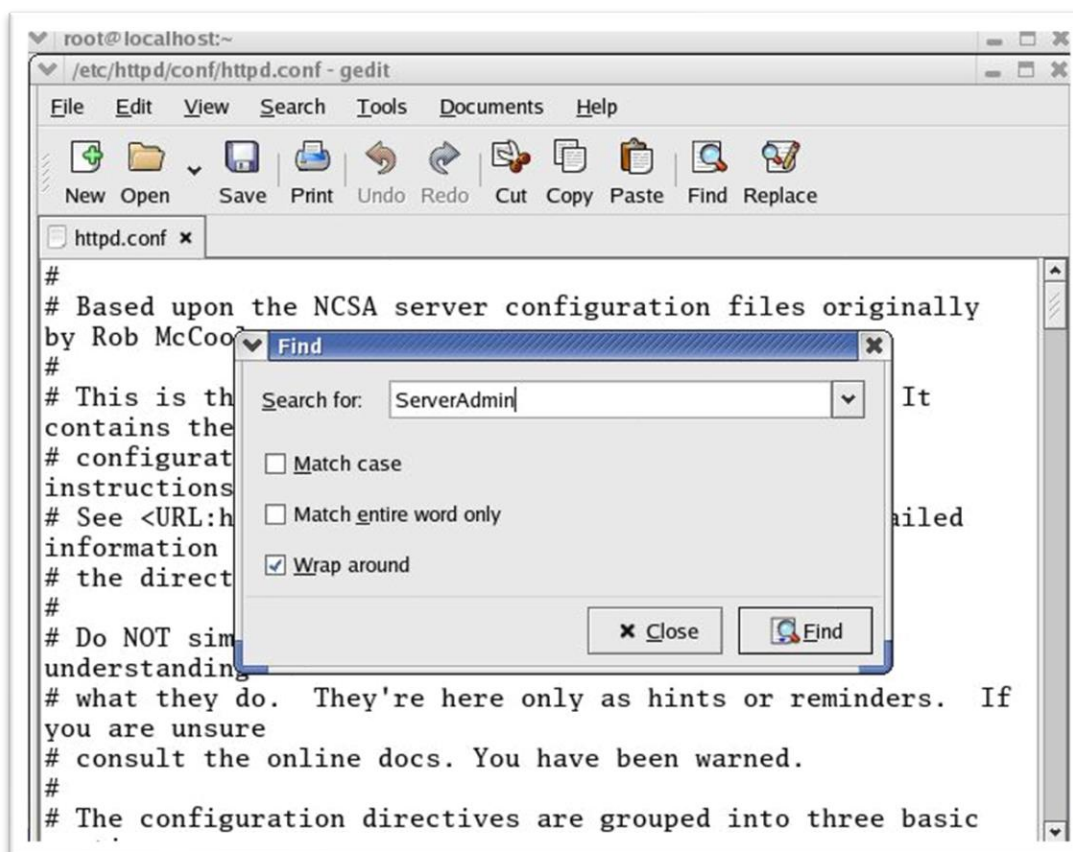
Testing the Apache Web Server

Once you've started the Apache web server, you should test it to see if it's working properly. To do that, we'll use a web browser to request a web page from our server!. There's a page provided by default for this purpose, and you can request it via the URL <http://localhost>. So, launch a web browser (Main Menu | Internet | Mozilla Web Browser), and type this URL into the address box:

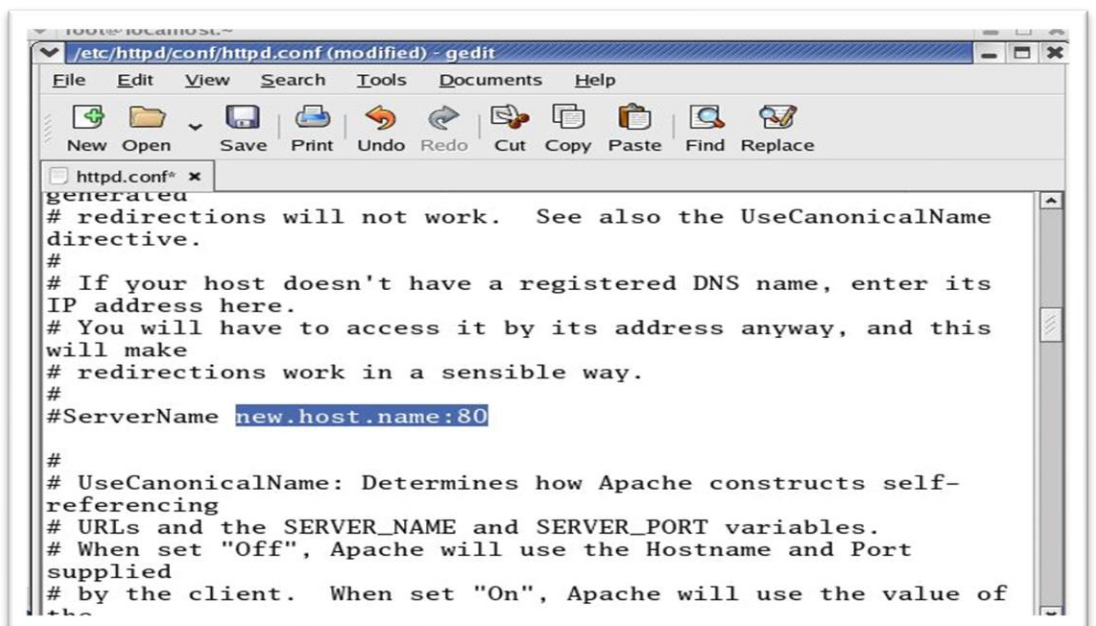
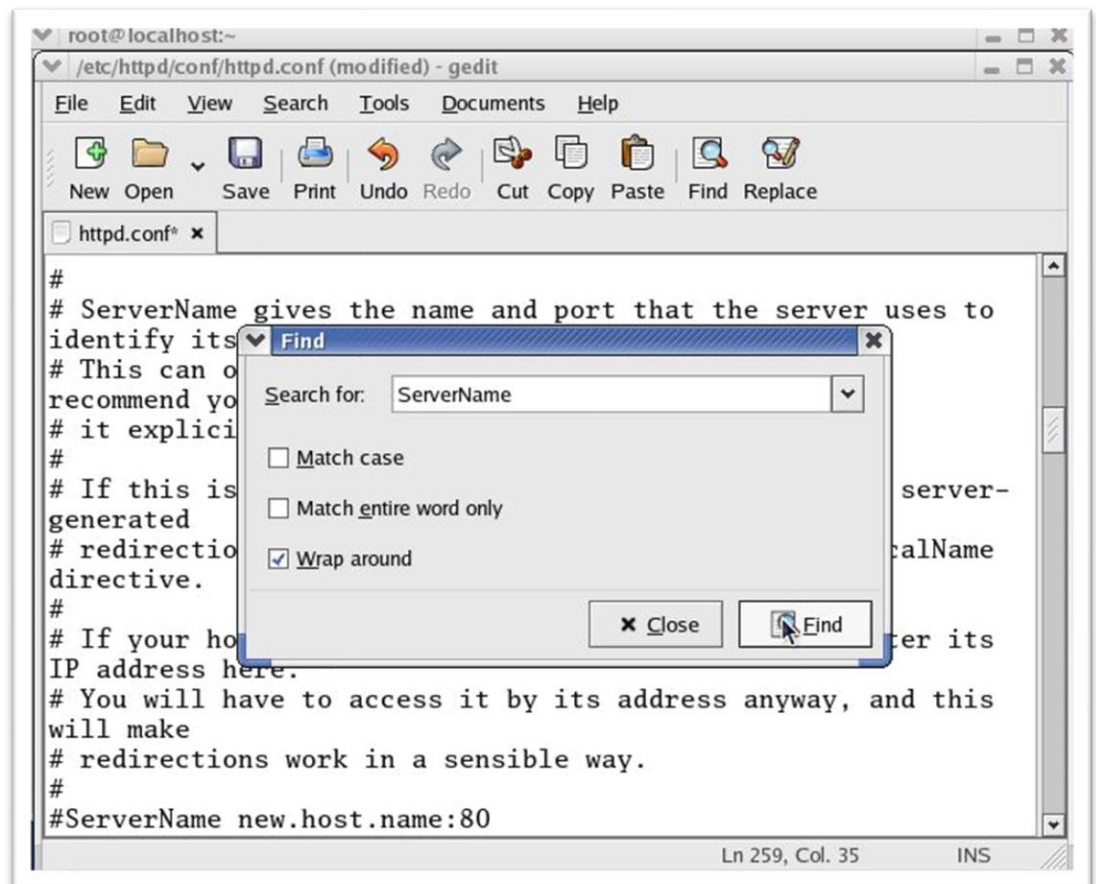


Configuring your Web Server

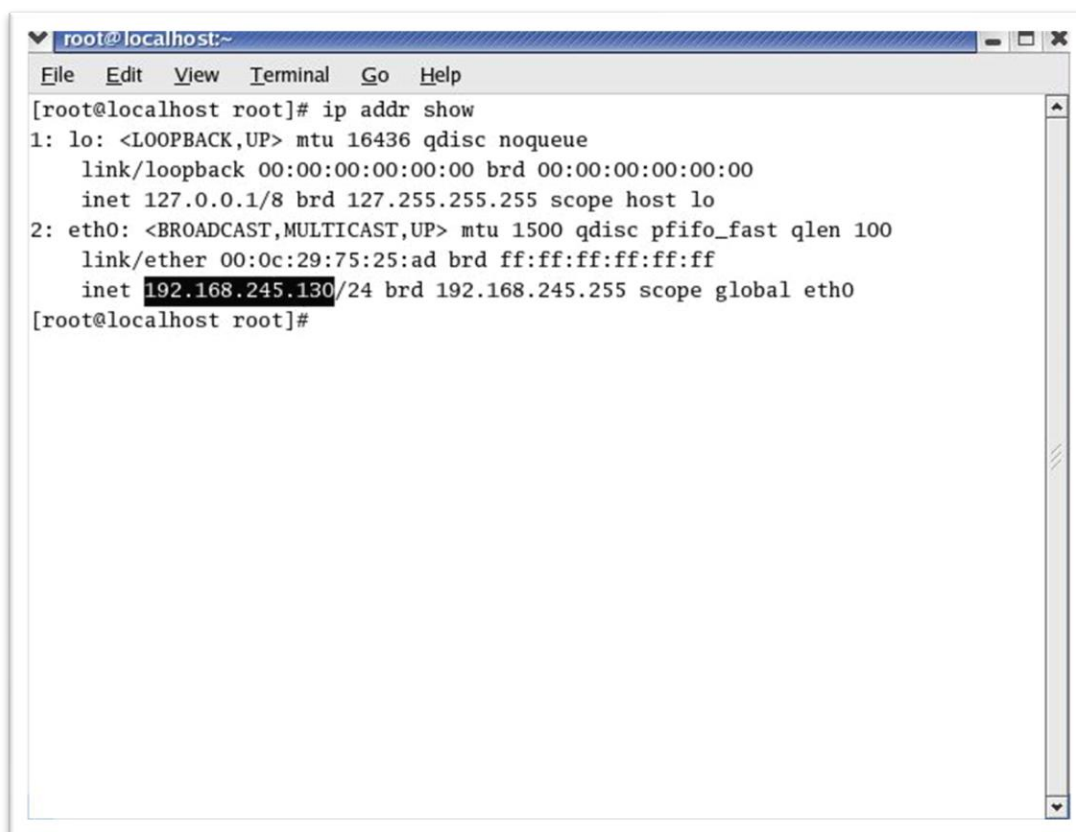
Launch the gedit text editor (by selecting Main Menu | Accessories | Text Editor). Use it to open the file `/etc/httpd/conf/httpd.conf`. Select Search | Find and use the resulting dialog to find the word Server Admin in the file. The first occurrence should be the Server Admin directive. Write your own email id here in the place of `root@localhost`.



Next thing we are going to search is ServerName. So in front of ServerName, new.host.name:80 is written. Here, we need to write the IP address of our machine.



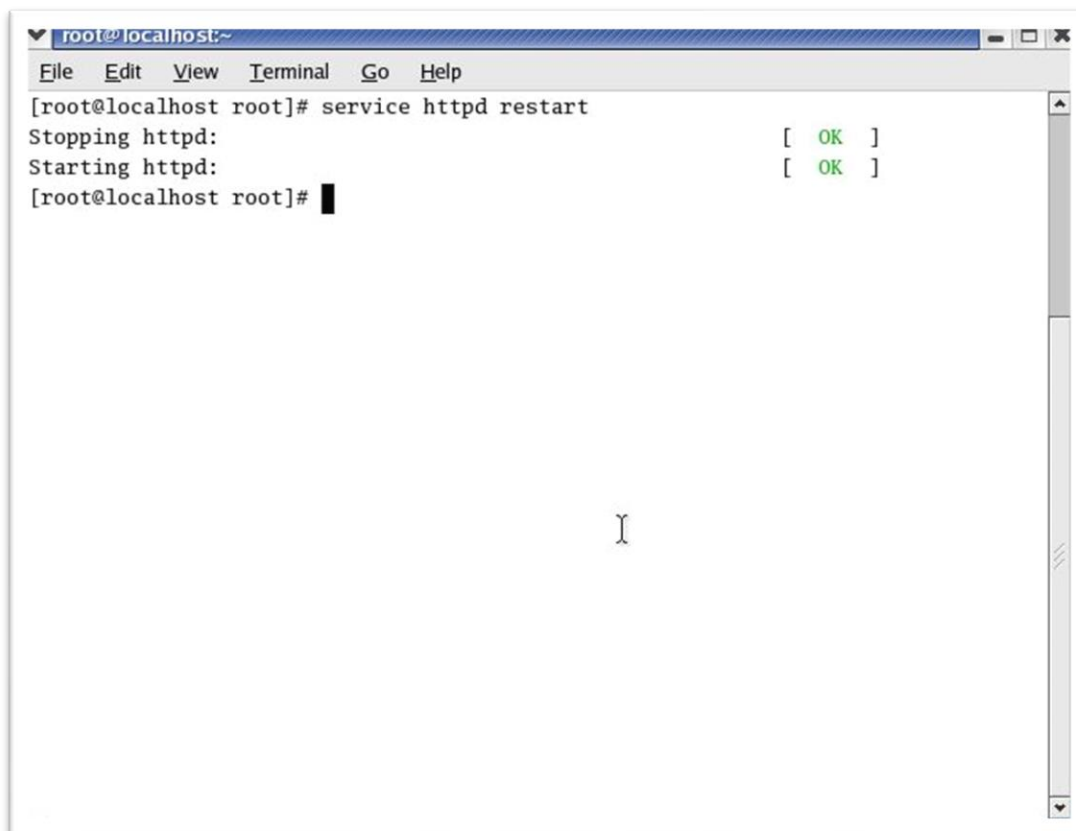
By giving the command `ip addr show`, it will show us the ip address of the machine. So, in front of ServerName write the ip address of the machine.

A terminal window titled 'root@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The command 'ip addr show' has been executed, displaying details for the loopback interface 'lo' and the ethernet interface 'eth0'. The IP address '192.168.245.130' for 'eth0' is highlighted with a black background.

```
root@localhost:~  
File Edit View Terminal Go Help  
[root@localhost root]# ip addr show  
1: lo: <LOOPBACK,UP> mtu 16436 qdisc noqueue  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 brd 127.255.255.255 scope host lo  
2: eth0: <BROADCAST,MULTICAST,UP> mtu 1500 qdisc pfifo_fast qlen 100  
    link/ether 00:0c:29:75:25:ad brd ff:ff:ff:ff:ff:ff  
    inet 192.168.245.130/24 brd 192.168.245.255 scope global eth0  
[root@localhost root]#
```

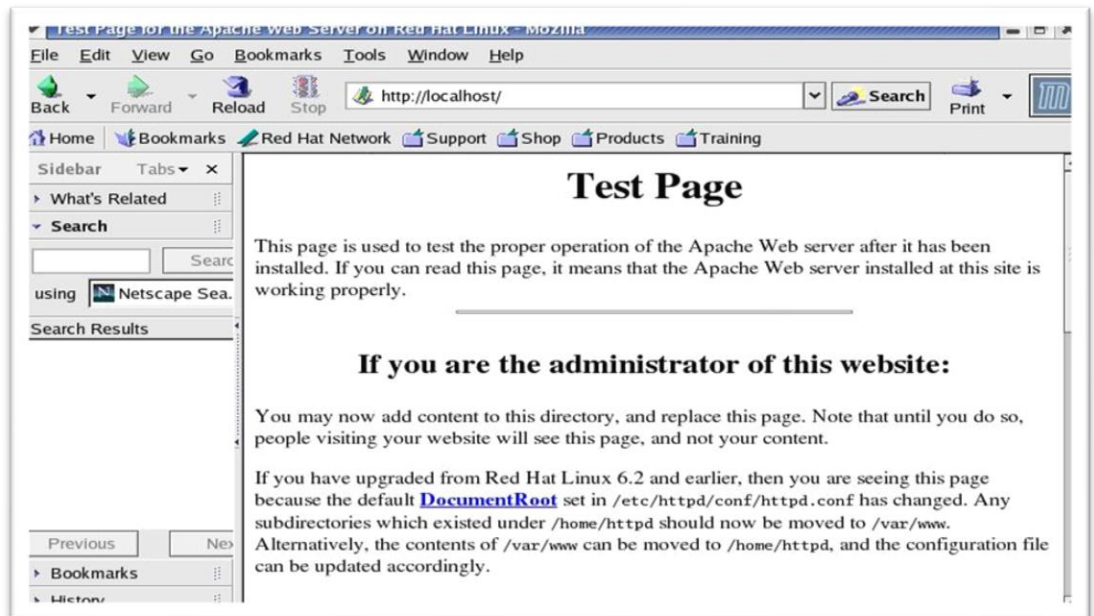
Configuring your Web Server

Again we need to restart the httpd service by writing service httpd restart.

A terminal window titled 'root@localhost:~' with a menu bar (File, Edit, View, Terminal, Go, Help). The command 'service httpd restart' has been executed, showing the process of stopping and starting the httpd service, both of which completed successfully as indicated by '[OK]'.

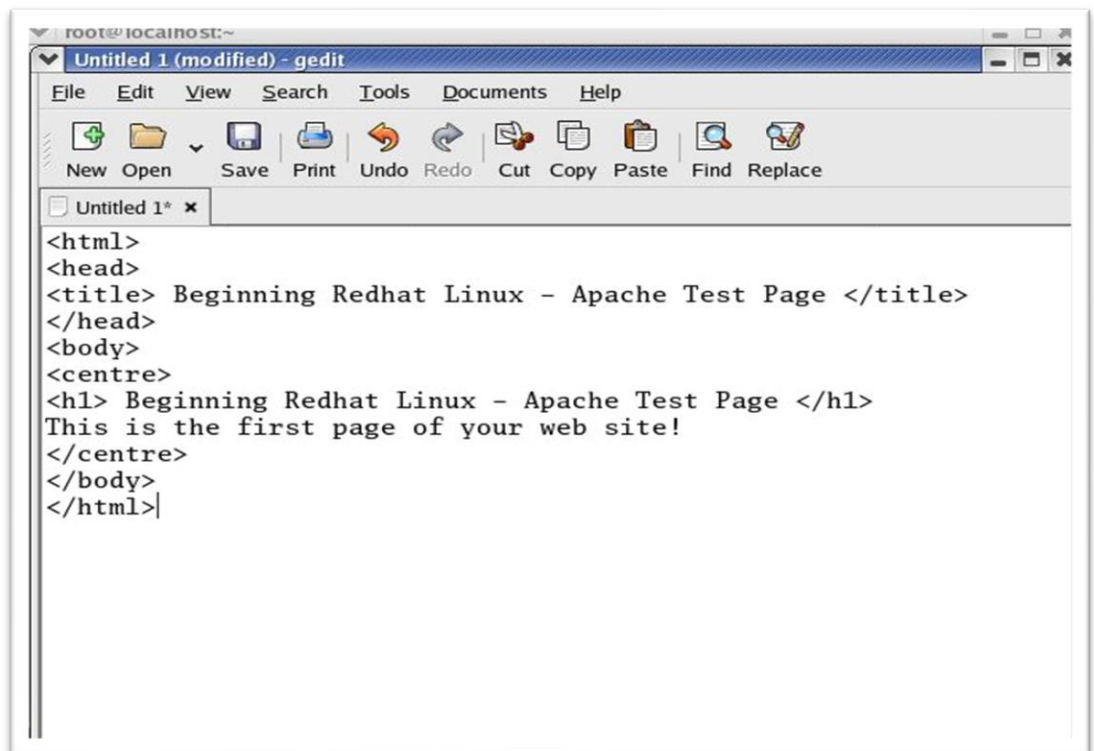
```
root@localhost:~  
File Edit View Terminal Go Help  
[root@localhost root]# service httpd restart  
Stopping httpd:          [ OK ]  
Starting httpd:          [ OK ]  
[root@localhost root]#
```

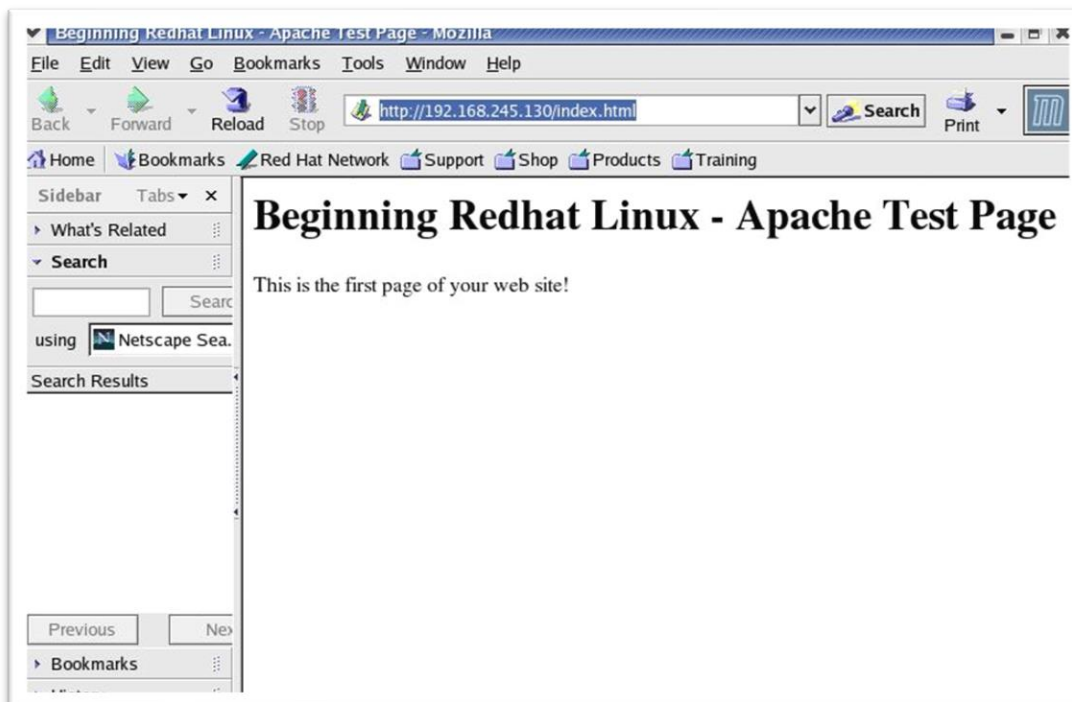
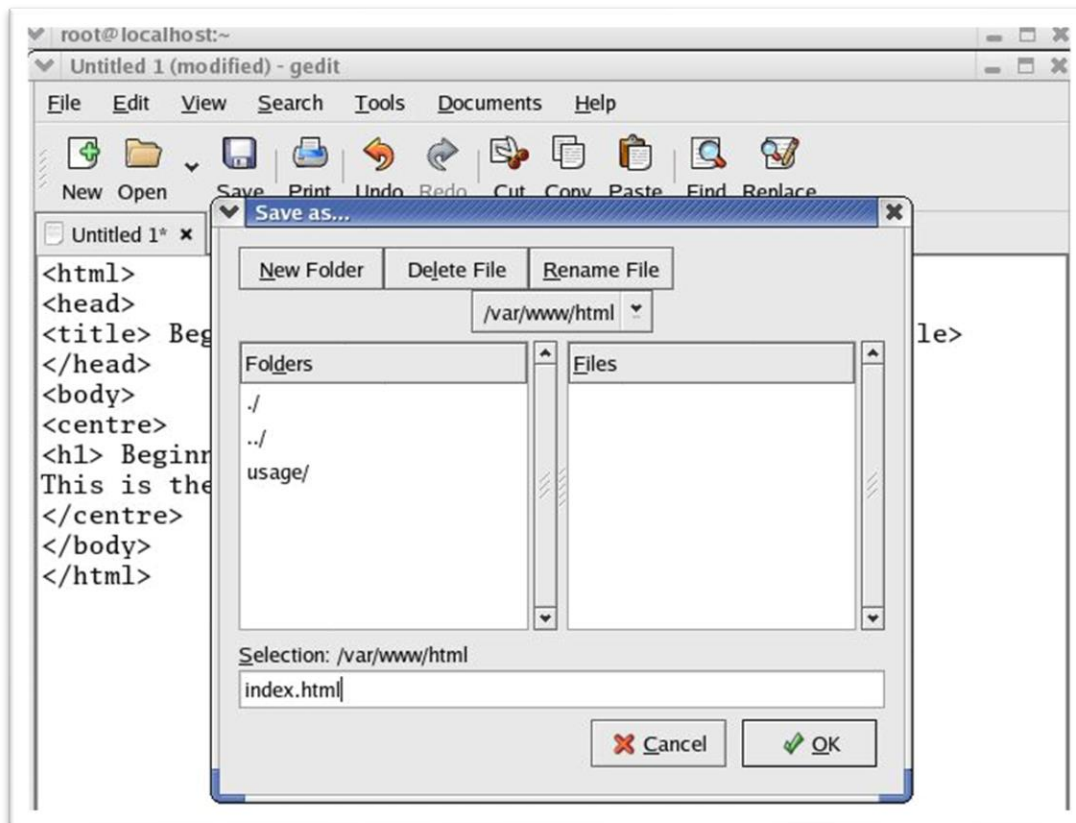
Once this is done. Again open `http://localhost`



Setting up Your First Web Page

- This involves creating a simple HTML web page, and saving it to a location on the hard disk that is used by the web server to store published web pages.
- Then, when a user requests the page, the web server will be able to respond by retrieving it from this location and sending it to the requestor.
- Launch an editor (gedit).





Summary

- A web server is required When we want to publish web pages on the Internet or on an intranet.

- The standard protocol for internet is Hypertext Transfer Protocol (HTTP).
- All web servers listen for HTTP requests on port 80.
- Apache's popularity is due not only of its open source pedigree, but also to its highly competitive levels of performance, functionality, stability, flexibility, and security.
- There are two ways to start RPM's graphical interface: One is: Main Menu | System Settings | Add/Remove Applications and another is \$ redhat-config-packages.
- The important package httpd_manual contains the documentation for the Apache web server. After installation, you can access this documentation from the command line by typing man httpd.
- The /etc/httpd/httpd.conf file is Apache's main configuration file.
- A web server is an application that does two things: It listens for page requests and when it receives a page request, it examines the request and responds with the page that was requested.

Keywords

- **Web server:** web server performs is to display the content of the website, which it does by storing, then processing, and eventually delivering the webpages to the user who has requested it.
- **Static Web Server:** The static content is the one that is fixed. The static web server contains HTTP software and computer. This is static because the server sends hosted files as is present to the browser.
- **Dynamic Web Server:** The dynamic web browser will have the webserver and the software like the database and the applications server. This is dynamic because the application server is used to update the files hosted before these are sent to the browser.
- **Apache Web Server:** Apache is a modular server – the core server provides the basic functionality, with extended features available in various modules. This makes it very flexible and easy to configure, because you need to configure only the modules you need.
- **HTTP:** Hypertext Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML. It was designed for communication between web browsers and web servers, but it can also be used for other purposes.

Self Assessment

1. When we are talking about the web servers, how many packages are available?
 - A. 15
 - B. 16
 - C. 17
 - D. 18
2. Which package contains the documentation of httpd web server?
 - A. httpd-manual
 - B. hwcrypto
 - C. php
 - D. php-image

3. We can install the web server package if we have logged in as
 - A. Root
 - B. Normal user
 - C. Either root or normal user
 - D. None of the above

4. When we are installing web server, which service should be started from service configuration dialog?
 - A. echo
 - B. httpd
 - C. cups
 - D. None of the above

5. The configuration file is modified by searching
 - A. ServerAdmin
 - B. ServerName
 - C. Both of the above
 - D. None of the above

6. Where do we provide the ip address of machine when the modification of configuration file is done?
 - A. ServerAdmin
 - B. ServerName
 - C. Both of the above
 - D. None of the above

7. The important packages for Apache web server are
 - A. httpd-manual
 - B. mod_ssl
 - C. php
 - D. All of the above mentioned

8. Which command is used to check the ip address of the Linux machine?
 - A. ip addr show
 - B. internet address show
 - C. ip address show
 - D. None of the above mentioned

9. Why do we require web server on a Linux machine?
 - A. To test the site that is under development
 - B. To test a private website only available in private network
 - C. Both of the above
 - D. None of the above

10. With which way we can open RPM's graphical interface?
 - A. Main Menu | System Settings | Add/Remove Applications
 - B. `$ redhat-config-packages`
 - C. Either of these mentioned
 - D. None of the above

11. HTTP stands for
 - A. Host text transfer protocol
 - B. Hypertext transfer protocol
 - C. High text transfer protocol
 - D. None of the above

12. The commercial web servers are
 - A. Zeus
 - B. Microsoft
 - C. SunOne
 - D. All of the mentioned

13. Which of these features belongs to Apache web server?
 - A. Portability
 - B. Scalability
 - C. Security
 - D. All of the above mentioned

14. What is the task of a web server?
 - A. Listens for a page request.
 - B. Examines and responds with the page requested
 - C. Both of the above
 - D. None of the above

15. Which one is the standard protocol of internet?
 - A. HTTP
 - B. PTTH
 - C. PTHT
 - D. None of the above

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. A | 4. B | 5. C |
| 6. B | 7. D | 8. A | 9. C | 10. C |
| 11. B | 12. D | 13. D | 14. C | 15. A |

Review Questions:

1. What is a web server? What is the need of installing web server on RedHat Linux machine? Give some examples of commercial and open source web servers.
2. What is Apache web server? Write some features of it in detail.
3. Write in detail how can we install Apache in RedHat Linux machine.
4. What are the important packages and configuration files of Apache web service. Give details.
5. How can we start and test Apache web server? Write in detail about its configuration.

**Further Readings**

Sandip Bhattacharya, Pancrazio De Mauro, Shishir Gundavaram, Mark Mamone, Kalip Sharma, Deepak Thomas, and Simon Whiting, Beginning Red Hat Linux 9, Wiley Publishing, Inc.

**Web Links**

<https://www.sumologic.com/blog/apache-web-server-introduction/>